OPEN-SOURCE TOOL · RUNS ON YOUR MACHINE · YOUR DATA STAYS YOURS

AI Job Search Assistant Install & Run Guide

The tool that does the heavy lifting of a job search for you - finding matching roles every day, checking each posting is still open, and building a resume tailored to that exact role - so your only job is to review and decide.

1. It never invents anything.
Tailoring rewords and reorders what is already in your master resume, and surfaces a job's keywords *only* where your resume already proves them. No new skills, projects, employers, dates or numbers. A programmatic check rejects any tailored resume that adds a skill your master resume does not contain.

2. It never submits anything.
It can open an application form already filled, with the tailored resume attached - then it stops, before Submit. You read the form, you click Submit, you mark it Applied.

Repository
github.com/ananyajoshi0203-cpu/ai-job-search-assistant

You need
Python 3.10 or newer + your own AI API key

Runs
Locally at http://localhost:8765 - no account, no upload

Your data
SQLite tracker + tailored/ folder on your disk. Persists when you close the app.

Start here - the 2-minute version

```
git clone https://github.com/ananyajoshi0203-cpu/ai-job-search-assistant.git
cd ai-job-search-assistant
# macOS: double-click run.command in the folder
# Any system, after the manual install on page 3:
jobhunt app
```

That opens the app in your browser. From there, everything is buttons - **Run crawl** **Tailor Found roles** **All roles** **Export .xlsx** - and on each role card, **Apply** and **No**. If you have never run a tool like this before, try the built-in fictional sample profile first (page 3). It costs nothing and risks nothing.

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1 What it does - the heavy lifting, step by step

Job hunting is mostly repetitive work: searching the same boards and rewriting the same resume ten slightly different ways. Configure this tool once with your real resume and target roles; after that, one run does all of the following.

1. Crawl
Searches your configured sources for roles matching your profile.

2. Dedupe
Checks every result against your tracker. Anything already seen - and anything you marked Skipped - never comes back.

3. Verify
Opens each new posting and confirms it is still live. Closed roles are recorded as Closed and never tailored.

4. Tailor
Builds a separate resume for that one role and drafts application answers from it.

5. Review
Presents each role as a card: job link, tailored resume, shared keywords, draft answers, Apply / No.

How tailoring works - rewording, not reinventing

For each role, the tool finds the **shared keywords**: skills the job asks for *and* your resume already shows. Only those may be emphasised. Tailoring can:

- **Reorder** sections and bullet points so the experience most relevant to this role comes first.
- **Reword** your existing bullets using the job description's own phrasing - for work you genuinely did.
- **Highlight** the required skills, tools and keywords that are already true for you, so a recruiter (and an applicant-tracking system) sees the match immediately.
- **Draft answers** to application questions, drawn only from the tailored resume.

It cannot add a skill, tool, project, employer, qualification, date or metric that is not in your master resume. A **no-invention check** then scans the output; if a skill slipped in that the master does not contain, that resume is rejected, not shown. Your master file is never overwritten - every role gets its own file in `tailored/`.

Where it looks

Source	How it works	Default
Greenhouse, Lever, Ashby	Official public APIs of the boards companies post jobs on. You add the board tokens for companies you care about.	On - most reliable
LinkedIn	Public, logged-out job search pages	On
Wellfound	Public search pages	On
Jobright	Public search pages	On

Read this before you crawl: LinkedIn, Wellfound and Jobright restrict automated access in their Terms of Service. Their adapters use public pages only - no login, no CAPTCHA bypass - with strict rate limits, and stop if blocked. Using them is your decision and your responsibility. In the config you can set `enabled: false` for any of them; the Greenhouse / Lever / Ashby APIs are the terms-clean path.

Statuses - so you always know where a role stands

Found → Ready to review → Applied plus Skipped (your No - permanent) and Closed (posting no longer open).

2 Install and run

Before you start

You need **Python 3.10 or newer** (`python3 --version` to check; if missing, install from **python.org/downloads**). You will also need an AI API key for tailoring - but you can crawl and explore first without one.

The easiest way

macOS: clone the repo and **double-click** **`run.command`** in the folder. It sets up the venv if needed, installs the tool, starts the app and opens your browser.

Any system (after a manual install):

```
jobhunt app
```

Starts the app and opens http://localhost:8765. Ctrl+C stops it.

Manual install - the flow that works

This sequence avoids the two failures almost everyone hits first: no `pip` command on macOS, and the "externally-managed environment" error. The virtual environment (venv) gives the project its own isolated Python.

```
git clone https://github.com/ananyajoshi0203-cpu/ai-job-search-assistant.git
cd ai-job-search-assistant
python3 -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\activate
python -m pip install -e .
jobhunt app
```

- **`jobhunt` is one word** - no space. `job hunt` will always say "command not found".
- **Activate the venv in every new terminal window** (`source .venv/bin/activate`); your prompt should start with `(.venv)` . Forgetting this is the number-one cause of "command not found".
- **Never type bare `pip`** - on macOS it often does not exist. Always `python -m pip` inside the venv.
- **Fallback if the command never registers:** from the repo folder, run `PYTHONPATH=src python3 -m jobhunt.cli crawl` (add your usual `--config` / `--db` flags before `crawl`).

Try the sample profile first (recommended)

The repo ships a fictional student - **Alex Rivera** - with a sample resume and config, so you can see the whole flow before using your own details.

```
jobhunt --config sample/sample_config.yaml --db data/demo.db crawl
jobhunt --db data/demo.db review
```

A crawl takes a little while - it verifies each posting is still open, and a slow source can take around 30 seconds before the crawl logs it and moves on. That is normal. "Found 10" in the final stats means ten verified roles were saved.

One config, one database - for every command. `--config` (default `config.yaml`) and `--db` (default `data/tracker.db`) go *before* the subcommand and must match across crawl, tailor and review - otherwise a command quietly looks at a different, possibly empty, tracker. Quick check: whichever `--db` makes `jobhunt list` show your roles is the right one.

3 Make it yours

Everything personal lives in one file. Copy the example, edit it, and keep it - the real `config.yaml`, your resume, your tracker and your tailored resumes are all git-ignored, so they never get committed or shared by accident.

```
cp config.example.yaml config.yaml
export OPENAI_API_KEY=your-key-here     # Windows: set OPENAI_API_KEY=your-key-here
jobhunt init                            # creates data/tracker.db
```

Setting	What it is	Change it to
`profile.name`	Your name, used in drafts	Your real name
`profile.roles`	Job titles to hunt for	e.g. "AI Engineer", "Machine Learning Engineer" - your target titles
`profile.locations`	Where roles must be	Your cities / regions / "Remote"
`profile.keywords`	Search keywords	Your field's terms - these drive the crawl
`profile.email_for_applications`	Email pre-filled on forms	The inbox you actually check
`profile.master_resume_path`	Your one true resume (Markdown)	Point it at your file - see below
`llm.model` / `llm.base_url`	Which AI does the tailoring	Default is gpt-4o-mini via OpenAI; any OpenAI-compatible endpoint works
`sources.greenhouse / lever / ashby`	Company job boards to watch	Add board tokens: the word after the domain, e.g. boards.greenhouse.io/**gleanwork**
`sources.linkedin / wellfound / jobright`	Aggregator searches	Your query + location, or `enabled: false` to switch one off
`tailoring.enabled`	Tailor during crawl, or stop at Found	`true` once your key is set; `false` to collect roles first

3 Make it yours - your resume and your key

Your master resume - the source of truth

Replace the sample resume path with your own Markdown resume. Write it once, write it completely and truthfully - every skill, project, employer, date and number you might ever want in an application must be in here, because tailoring can only use what is in this file. If a real achievement is missing from the master, it will never appear in a tailored version. That limit is deliberate: it is what makes invention impossible.

Crawled before you set your key? Use Tailor, not another crawl

Roles only become review cards (*Ready to review*) after tailoring, and tailoring needs the API key. If you crawled first, your roles sit at *Found* - and re-running crawl will not fix it, because known roles are skipped as duplicates. Instead:

```
export OPENAI_API_KEY=your-key-here
jobhunt --config sample/sample_config.yaml tailor
jobhunt review
```

Use the same `--config` / `--db` pair you crawled with. Tailor re-checks each Found role is still open, builds its resume and draft answers, and moves it to Ready to review. The same action is a button in the app: **Tailor Found roles**.

One-time setup for form pre-fill

```
python -m pip install -e ".[prefill]"
playwright install chromium
```

Or click **Set up pre-fill (one-time)** in the app toolbar and let it finish. Without this extra, everything else still works - Apply will simply tell you pre-fill is not set up yet.

4 Daily use from the page

Once the app is running, you should not need the terminal again. The page at `http://localhost:8765` is the whole tool.

The toolbar

Button	What happens
Run crawl	Runs a full crawl with the config the app was started with: search, dedupe, verify, and tailor if your key is set.
Tailor Found roles	Takes roles already sitting at Found, re-checks each is still open, builds a tailored resume and draft answers for each, and moves them to Ready to review.
All roles	Every role in the tracker, with status pills and filters for Found, Ready, Applied, Skipped and Closed.
Export .xlsx	Downloads the tracker as a spreadsheet you can open, sort and share.
Review	Back to the Ready to review cards.
Set up pre-fill (one-time)	Installs the browser extra used by pre-fill. Click once, let it finish, then use Pre-fill.

Working a card

1. **Open the job.** The role title links to the full job description on the employer's site. Read it first.
2. **Open the resume link.** It now opens a viewer inside the app showing the exact tailored resume built for this application, with its draft answers underneath. Read it like a recruiter would.
3. **Approve the resume.** Click **Approve this resume** in the viewer. The card then shows a "Resume approved" marker, so you can see at a glance which ones you have signed off on.
4. **Click Apply / Pre-fill form.** Where pre-fill is set up, this opens the employer's form with your details filled and the tailored resume converted to .docx for upload. Check every field and the attached file name on the employer's page - if a form uses a custom uploader, attach the .docx from the `tailored/` folder yourself.
5. **Submit it yourself** in that employer window, if everything is right. The tool never clicks Submit for you.
6. **Click Mark as Applied** back on the card. The role moves to Applied with today's date, the Applied counter climbs, and it appears under the Applied filter.
7. **Or click No.** The role is marked Skipped and never comes back in a future crawl.

The top counters are links

Click **Ready to review**, **Applied**, **Skipped**, **Found** or **Closed** to jump straight to that filtered list.

Your data persists

Yes - all of it. Roles, statuses, notes, approval markers and dates live in the SQLite file (`data/tracker.db` by default), and tailored resumes live in `tailored/`. Every click writes immediately. Kill the app, close the laptop, restart tomorrow: everything is exactly where you left it. Back up those two places and you have backed up the tool.

Run it daily (optional)

A cron line such as `0 8 * * * cd ai-job-search-assistant && jobhunt crawl` runs a morning crawl, or use the GitHub Actions workflow in `.github/workflows/daily-crawl.yml` with your key as a secret.

5 Troubleshooting

What you see	Fix
`command not found: jobhunt`	Activate the venv in this terminal (`source .venv/bin/activate` - Windows: `.venv\Scripts\activate`) and check spelling: `jobhunt` is one word. If it still fails, the install never finished - run `python -m pip install -e .` inside the venv.
`command not found: pip`	Normal on macOS. Never use bare `pip` - use `python -m pip` inside the venv.
`externally-managed-environment`	You installed outside the venv. Create and activate the venv first (page 3), then install again.
`FileNotFoundError: 'config.yaml'`	You ran a command with no `--config` before creating `config.yaml`. Either create it (page 4) or pass the sample config explicitly.
Review page shows all zeros / no cards	You are looking at a different database. Stop the app (Ctrl+C), run `jobhunt list` with candidate `--db` values until your roles appear, then start review/app with that same `--db`.
Roles stuck at Found, no cards	Tailoring has not run - usually no API key was set during crawl. Set the key and run `jobhunt tailor` (or the Tailor Found roles button) with the same flags. Do not re-crawl.
`invalid choice: 'tailor'` (or another command)	Your clone is behind the repo. Run `git pull` - no reinstall needed, the editable install reads the new source live. Then restart the app.
`"ready": 0, "errors": []` from tailor	That database has no Found roles in it. It is almost always the wrong `--db` - see the zeros fix above.
Pre-fill says the browser extra is needed	Click **Set up pre-fill (one-time)** in the toolbar and let it finish, or run `python -m pip install -e ".[prefill]"` then `playwright install chromium`. Restart the app afterwards if it was already running.
Form opens but resume is not attached	Tailored resumes start as Markdown; pre-fill converts them to .docx for upload, because job forms do not accept .md. Check the employer's form for the .docx file name. If that board uses a custom drag-and-drop uploader, attach the .docx from `tailored/` yourself.
A JavaScript error mentioning "undefined" / "Cannot read properties"	That message comes from a web page, not from this tool's Python code. Note the full text after "reading" and which window showed it - the app page, or the employer's form - before changing anything.
Old buttons or old behaviour after an update	The running app still holds the old code in memory. `git pull`, then Ctrl+C and restart the app.

The command reference (if you prefer the terminal)

```
jobhunt init        # create the tracker
jobhunt crawl       # search, dedupe, verify, tailor
jobhunt tailor      # tailor roles already at Found
jobhunt list        # list tracker rows (--status to filter)
jobhunt review      # review page on http://localhost:8765
jobhunt app         # review page + opens your browser
jobhunt export --out tracker.xlsx
jobhunt skip <job_id>
jobhunt prefill --job-id <id> --answers <file.json>
```

Global flags go before the subcommand: `jobhunt --config config.yaml --db data/tracker.db crawl`.

What's still in progress

You will need to apply yourself - the automation for applying is still in progress. What the tool does today is the heavy lifting around it: it finds and verifies your leads, maintains your status and leads in the tracker so nothing is lost or duplicated, and it does the resume tweaking for each role - rewording what you already have, surfacing the right keywords, and checking that nothing was invented. Apply and Pre-fill help inside that last step where an employer's form allows it, but the final application - reviewing the filled form, confirming the right resume is attached, and submitting - is yours to do, every time.